BAYMAX Experiment Notebook

This folder explains the experiments, rather than merely listing result filenames. Each numbered note covers one training experiment, inference experiment or clearly identified measurement/audit investigation. Multiple evaluations of the same checkpoint can live in its note; a 30-case diagnostic does not become a separate trained model.

Start with a question or follow the stages below. Numbers within folders are reading order, not invented run IDs or a guarantee of exact historical chronology. Every evaluation table links to its original JSON and records its saved date.

Where We Stand

The selected development reference is cleaned anchored SFT at two epochs. It is the reference for balanced routing, not the winner of every metric.

Candidate	Task successes / 150	Routing macro-F1	Invalid outputs / 150	Premature ACT on boundary cases
Anchored two-epoch SFT	44	0.514	2	31
DPO from that reference	53	0.376	5	56
2:1 SFT	29	0.477	8	22
DPO from 2:1 SFT	31	0.453	7	34
DPO with completed ACT rejections	46	0.482	7	45
Rank-16 two-epoch SFT	46	0.450	7	42
Lower-learning-rate two-epoch SFT	45	0.436	15	37
QKVO two-epoch SFT	52	0.437	14	40

Those are measured development results, not evidence of a robust overall win. First DPO has the highest task completion but substantially worse abstention. Completing its missing rejections recovers some balance, yet remains below original SFT macro-F1. Rank 16 also worsened routing and validity despite two additional task successes. Lower learning rate did not improve balance either. QKVO completes more tasks but has worse routing and output validity: retain it as a task-completion candidate, not the balanced reference. A's original rank-8 recipe remains the reference. No model switch or new training method was implemented by this documentation task.

The full recipes, matrices, sources and caveats are in the individual notes. For the most recent sequence, read the two-epoch reference, four-epoch trial, matched contrasts, 2:1 trial, and the three DPO notes in that order.

How an Experiment Flows

1. Author or import training examples under the general policy and tool contracts.
2. Protect source train/validation/test membership and contrast/content groups.
3. Mix and sample a deterministic raw training cap.
4. Render the chat template, tokenize, mask non-assistant labels and truncate.
5. Inspect the usable artifact that actually reaches the trainer.
6. Train a new versioned adapter.
7. Run the chosen adapter on a separate evaluation suite through fake adapters.
8. Read task outcome, routing, output validity and wording metrics separately.

Pipeline fixes at steps 2-5 or 7-8 can change the validity of the evidence without creating a better trained model. That distinction explains several apparent jumps in the history.

Vocabulary

Term	Meaning in these notes
ACT	Select supported productive tools when the request is safe and complete.
CLARIFY	Supported safe action needs missing/ambiguous information or required confirmation.
REFUSE	Action/capability is unsafe, unavailable, unsupported or out of scope.
Meta-tool	request_clarification(question) or refuse_request(reason); not an external mutation.
xLAM	External Salesforce function-calling dataset hosted on Hugging Face.
BAYMAX-control	Authored training data; "control" is the dataset name, not necessarily a baseline model.
Mixed training	xLAM plus authored BAYMAX examples.
SFT	Learn supplied correct assistant answers using target-token cross-entropy.
LoRA	Train small adapter weights rather than updating every base-model weight.
DPO	Learn to prefer a chosen answer over a rejected answer for the same prompt.
Anchored sampling	Keep reference examples/positions and replace only explicitly eligible rows.
Matched contrast	Closely related requests/states/tools with a genuine decision-changing difference.
Raw cap	Number selected before tokenization and assistant-label filtering.

Term	Meaning in these notes
Usable rows	Examples retaining trainable assistant labels after preprocessing.
Epoch	One pass over the usable training artifact; not one optimizer step.
Exact duplicate	Identical example content; not a test of semantic or template similarity.
Reference freeze	Record hashes to detect drift; it does not lock or copy files.

Metrics Are Different Questions

- **Task success:** did all scenario checks pass, including arguments and message criteria?
- **Routing macro-F1:** average ACT, CLARIFY and REFUSE F1 equally. F1 combines precision (correctness of predictions of that class) and recall (coverage of expected cases of that class). Invalid/unknown outputs miss their expected class.
- **Decision accuracy:** correctness among recorded known decisions. Read it with coverage; unknown outputs are not included in its conditional denominator.
- **Clarification/refusal accuracy:** required question/reason keyword matches. A correct label can still have an incomplete or misleading explanation.
- **Invalid-output rate:** strict parser rejections, separate from valid but inappropriate decisions or tool-execution failures.
- **Hallucination rate:** currently unavailable tool names, not every invented person, date or success claim. Invalid parsing can leave no accepted calls to count.
- **Latency/cost:** local reported API cost zero is not zero electricity, hardware or training cost. Replayed results inherit original inference timing.

Tables use ratios from zero to one, rounded to three decimals, unless they show explicit counts. `n/a` means unavailable or inapplicable, not zero.

Which Evaluation Suite?

Suite	Size	Purpose
v1	50	Original benchmark, used by early trials.
Combined v1 + v2	150	Established development comparison. Current decision support is ACT 81, CLARIFY 40, REFUSE 29.
Template shift	30	Separate wording/structure diagnostic: ten cases per decision.
Refusal subset	7	Small single-class diagnostic, not balanced overall evaluation.
Smoke subsets	1-4	Sanity checks; not estimates of full benchmark quality.

Historical policy-era fixtures/scoring differ from current measurements. Do not combine the template-shift and established-suite rates into one score. Their repeatedly inspected status means neither is an untouched final benchmark.

Early Training

The first external-only, synthetic-only and mixed trials. Historical provenance is incomplete.

- External Tool Data Only: First Local Fine-Tune
- BAYMAX Control Data Only
- Control Data With Contrast Pairs
- Control Data With Targeted Refusals
- Original External and BAYMAX Training Mix
- Expanded Control Generator: Early Mixed Trial
- Weighted Control Examples: Early Oversampling Trial
- Clarification and Refusal as Explicit Meta-Tools

Format, Prompt and Sampling

Input alignment, weighting, unique-first caps and inference-only prompt changes.

- Aligning Training and Evaluation Formats
- Extra Clarification and Refusal Repeats
- Legacy Boundary-Weighting Trial
- Fixed Unique Boundary Pool at 1,000 Rows
- Increasing the Cap to 1,500 Without Repeating Boundaries
- First Policy Prompt: Evaluation Without Retraining
- Expanded Policy Prompt: Evaluation Without Retraining
- Returning to the Recorded Training Prompt

Data Quality and Class Mix

Changes in class quantities, partition integrity, policy, credentials and target compatibility.

- One Hundred ACT, CLARIFY and REFUSE Examples
- Early Balanced BAYMAX Data at Two Epochs
- Schema-Diverse Control Examples
- Increasing Productive BAYMAX Examples to Two Hundred
- Productive Midpoint: One Hundred Fifty BAYMAX ACT Examples
- Content-Group Splits With Seventy Percent ACT
- Corrected Policy, Credential Filtering and Representative Validation
- Corrected Policy With Two Hundred Productive Examples

- Target-Contract Cleaning With Resampled External Rows
- Target-Contract Cleaning With Anchored External Rows

Training Duration and Matched Contrasts

Fresh one/two/four-epoch comparisons, matched groups and the 2:1 SFT ratio.

- Anchored Data at Two Epochs: Selected Reference
- Anchored Data at Four Epochs
- Matched Contrasts at Fixed Class Counts
- Two-to-One Productive-to-Boundary SFT Ratio

SFT Hyperparameters

Controlled adapter/training changes from the selected SFT reference. B, C and E are completed. Increasing rank, lowering the learning rate and adding attention projections did not beat A's balanced routing. E improved task completion, with output-contract failures documented separately.

- Experiment B: Rank 16 at the Fixed Two-Epoch SFT Recipe
- Experiment C: Lower Learning Rate at Two Epochs
- Experiment E: LoRA on All Four Attention Projections

Preference Training

The original DPO stage, alternative SFT initialization, and completed ACT rejections.

- DPO From the Selected Two-Epoch SFT Reference
- DPO From the Two-to-One SFT Model
- Completing Missing Productive Rejections for Boundary Decisions

Measurement and Data Audits

Baseline interpretation and engineering corrections. These are not new model-training wins.

- Untrained Qwen: What the Baseline Actually Measures
- Effective Token Distribution and Training-Cap Audit
- Strict Parser Replay: Correcting Measurement, Not Training
- Training Data Contract and Policy Review
- Decision Metrics Separate From Explanation Wording
- Clarification Output Contract: Target Audit and Failure Diagnostics

Rules for Reading Comparisons

1. Compare the same scenario suite, prompt, generation settings, parser and scorer.
2. Distinguish a new trained checkpoint from an inference-only prompt test or strict replay of saved generations.
3. Check actual usable rows and tokens. A 1,000-row raw cap can deliver 974 rows.
4. A 200/100/100 BAYMAX quota plus 600 xLAM ACT examples is 800/100/100 overall, not a balanced three-class mixture. Multi-tool sequences still count as ACT.
5. Matching broad quantities does not guarantee matching domain/subtype coverage.
6. A fresh four-epoch run is not resuming a saved two-epoch adapter. A changed cosine horizon also changes per-step learning rates.
7. A 705-pair and an 800-pair DPO run at one epoch have different training exposure.
8. Single-seed changes are observations, not statistical proof or established causes.
9. Do not turn an inspected evaluation failure into a fixture-specific training target.
10. Do not assign today's config defaults to an old checkpoint with missing provenance.

Reproducibility and Sources

The external source revision used by the pinned pipeline is `26d14ebfe18b1f7b524bd39b404b50af5dc97866`. The later matched training prompt is identified by `a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468`.

Configs can declare context 2,048/batch two/accumulation eight while owner low-VRAM commands actually use 1,024/one/16. Later notes state actual reported overrides. Earlier notes explicitly retain missing information instead of filling it by guess.

Original evidence remains in the result folder. That older catalog predates some recent runs; use these notes for their explanations. FILE_RENAMES.json resolves historical names without changing result bytes. Config links identify recipes; generated model/data links are local-only and may not exist on another checkout.

Methodology remains defined by BENCHMARKING.md and SCORING.md. Detailed investigations remain in TRAINING_DATA_AUDIT.md and DPO_EXPERIMENT.md. Some older status paragraphs describe preparations that later completed; measured result artifacts take priority.

This notebook covers all 65 saved Phase 2 evaluation JSON files present on 2026-10-10, plus early external-only local results and supporting audits. No raw result, training example, evaluation target or model weight was changed. No long training/evaluation/audit command was run. These notes are not staged, committed or pushed; the owner's restriction on uploading Phase 2 documents remains.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.